Care Creations™

Hyaluronic Filling Spheres™
by **Beauty Creations**

BASF
The Chemical Company

Summary file

Hyaluronic Filling Spheres™	REFERENCE C00309		
Origin - Description	Dehydrated hyaluronic acid microspheres.		
Regulatory data			
INCI	Ethylhexyl Palmitate, Silica Dimethyl Silylate, Butylene Glycol, Pentylene Glycol, Sodium Hyaluronate, Caprylyl Glycol, Hexylene Glycol, Phenoxyethanol.		
Japanese Cosmetic Denomination	Parumichinsanokuchiru, Jimechiru Shiriruka Shirika, BG, Penchiren Gurikoru, Hiaruronsan Na, Kapuririrugurikori, Hekizhirengurikoru, Fenokishietanoru.		
CAS#	29806-73-3, 68611-44-9, 107-88-0, 5343-92-0, 9067-32-7, 1117-86-8, 107-41-5, 122-99-6.		
EINECS#	249-862-1, 271-893-4, 5343-92-0, 203-529-7, 232-678-0, 214-254-7, 203-489-0, 204-589-7.		
Preservatives/Additives	none		
Temporary specifications		-	95.0 - 100.0
	PHYSICAL ANALYSES		
	Appearance	-	Liquide
	Color	-	Translucent white
	Odor	-	Characteristic
	Viscosity (Brookfield RVDV-II +)	-(03;50)	(500-2 000) cps
	BACTERIOLOGICAL ANALYSES		
	total aerobic bacteria (30 °C)	-	< or = 100/g
	Pathogens : (S. Aureus; E. Coli; P.Aeruginosa; C.Albicans; A.Niger)	-	None
Formulation data			
Properties	• Smoothing deep wrinkles, • Long lasting hydration.		
Applications	• Anti-wrinkle skin care, • Mature skin eye treatment, • Deep wrinkle skin care (frown lines, forehead wrinkle, nasogenian folds), • Intense moisturizing skin care.		
Concentration of use	0.5 to 3%.		
Incorporation method	Pre-spread in the emulsion oily phase before making the emulsion. Do not use in aqueous gel or in an emulsion with a less than 20% oily phase.		
Patent	PERRIER E., HUC A., BUFFEVANT C. Polysaccharide wall microcapsules containing primary alcohol functions and compositions containing same. *(France: FR 2 688 422), (Europe: EP 0 630 287), (USA: US 5 562 924).*		

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Introduction

Hyaluronic Filling Spheres™ Hyaluronic acid microspheres: filling from biotechnology... A non-invasive alternative to surgery.

Thanks to their small size, dehydrated Hyaluronic Filling Spheres™ penetrate into skin folds of varying depth and trap water in the superficial layers of the skin.

The remarkable characteristic of hyaluronic acid, that acts as a molecular sponge, enables the spheres to swell and exert pressure in the direction of the surface, resulting in a smoothing effect on skin relief.

Hyaluronic Filling Spheres™, cosmetic filling product is based on using spheres of biotechnological origin that can provide volume to the superficial layers of the epidermis.

LAMARTINE (1790-1869) The lake:
O time, suspend your flight…

Opinion of the press

An unavoidable consequence of time is that the skin becomes thin, stretched and covered with blemishes. Connective fibers tend to agglutinate, the skin loses firmness and elasticity, fine lines appear on the surface and then deepen, ultimately giving rise to wrinkles.
Although aging of the skin is the visible traitor, this is merely one of the aspects of our body's aging… evidently aggravated by life style (alcohol, tobacco…) and exposure to sun. UV radiation (principally UVA and UVB) and the free radicals they produce accelerate skin aging.
Although biological age can theoretically be an objective criterion of human health, and by extension a means of predicting the course of the aging process, this is not true in practice. Biological, or functional age, and chronological age do not always coincide.

"Senescence is not a slope all of us descend at the same speed. It is in fact a series of irregular steps that some people go down faster than others». Our skin is the initial instrument of communication towards others and belies our apparent age. Every additional line at the corners of the eyes or lips is the reward of 2 or 3 additional years!

Faced with this, we have several possible strategies:

The first arm - the only tool available in fact - is simply … prevention. This process must be started in childhood, avoiding excessive exposure to the sun and using sunscreens.
Even though prevention is indispensable, it does not respond to all current needs …and above all it is often too late: the damage has been done and we are enter the long corridor of repair.

Research has thus accelerated in several directions and today's variety of «solutions» proposed is in the range of the most trivial to the most «interventional». Let us start with the «heavy artillery»: surgical treatment.

- Plastic surgery proposes re-firming tissues or filling in thinned zones - wrinkles - with two types of injectable filling products.
  The first are bovine collage biological products (almost pure type 1 collagen, whose peptide ends have been removed to render it antigenically inactive) or hyaluronic acid (fermentation product), that is also the most abundant glycosaminoglycan (GAG) in young people.
  The second are chemicals such as silicone, mixtures of bovine collagen and methyl methacrylate microparticles, but they are not without risk since they are a potential source of complications with varying severity.

- In parallel to these relatively «simple» techniques, surgery remains a possibility for implanting synthetic products such as textile fibers or gold thread, or even an autograft composed of dermis, fat or fascia (muscle insertion fibers) taken from the patient.
  It should be noted that autocollagen obtained from fat removed by liposuction can be reinjected only in the donor and even then, the results do not last indefinitely…

- Finally, it is difficult to end the discussion of plastic surgery techniques without talking about photothermolysis.
  In this field, «resurfacing laser» or «laser smoothing» uses the power of computers (pulsed scanner) to attack wrinkles precisely, much as smart bombs.
  The depth of wrinkle destruction is calculated automatically by the laser that produces a highly localized vaporization without damaging adjacent tissues.
  The procedure can be conducted under local anesthesia for isolated wrinkles or anesthetic premedication when the area to treat is larger (lips, eyelids). It is of course possible to combine this technique with a much more invasive surgical procedure: face lifting…

The claimed objective of many cosmetic products for many years has been to prevent the appearance of wrinkles and to reduce their impact.
The classification of «treatment» products involves varying principles. Some claim a «biological» action on the thickness and/or properties of the different skin layers.
Others act on more physicochemical aspects and produce a flattening of the skin surface.
Finally, other products act via optical effects that mask or simply render the wrinkle less apparent.

Between the two extremes, there is a gap … to fill!



Wrinkle filling care and cosmetics

What is a wrinkle?

A wrinkle can be defined as a rupture of elastic fibers and collagen fibers occurring in the epidermis and the dermis, combined with a quantitative reduction of dermal substances (glycosaminoglycans (GAGs), collagen…) that increases with age. Starting at the age of 30 on average, the first lines appear on the surface of the skin but easily disappear after the skin is gently stretched mechanically.

There are several types of wrinkles that are distinguished either by their location or their mechanism of formation, but all can be intensified by various intrinsic (hereditary, hormonal…) or extrinsic factors (sun, pollution, tobacco…). They are found primarily on the face (around the mouth, the eyes and on the forehead) and the hands, zones that are naturally exposed to the environment.

So-called expression wrinkles on the face are folds of varying depth that form when you knit your brows or smile.

They form along the lines of natural tension of the skin (Langer's line[1]) and no longer disappear when stretched since they arise from muscular contraction.

With increasing age, the fold remains the same and becomes definitive often along an orientation perpendicular to the underlying muscle layer. These folds are categorized as:

- forehead wrinkles: vertical or horizontal wrinkles on the forehead.
- frown lines, between the eyebrows.
- crow's feet wrinkles, at the contour of the eyes.
- naso-labial wrinkles around the lips.
- wrinkles of the nasolabial fold, that form after a movement of the cheek that descends and causes a fold starting from the nose and descending along the mouth.
  These wrinkles of the nasolabial fold are a good example of wrinkles due to ptosis, i.e. the downward displacement of subcutaneous muscles and the skin covering them. In plain language, these wrinkles appear when the skin loses its elasticity and is subjected to the force of gravity.

Expression wrinkles are wrinkles arising from repeated and frequent movements, requiring the activity of sub-conjonctival muscle masses in the same direction (mimics of expression), accompanied by muscular laxity. Depending on their location, the depth of these expression wrinkles can vary, but they are consistently more pronounced than other types of wrinkles (around the mouth or the neckline), that although clearly finer, are nevertheless a sign of advancing age.

Causative or aggravating agents are often environmental factors such as the sun, tobacco, alcohol, pollution or insufficient skin care.

Surgery: an effective but invasive response

The slackening of skin tissue or an excess of skin on the neck and cheeks is treated preferentially by surgery. In the case of expression wrinkles, the strategy involves relaxing the muscles, in particular by the use of botulinum toxin, or Botox.

Concerning the wrinkles arising from the sagging of the lower face (jowls, neck), blepharoplasties (plastic surgery for the eyelids), etc…, the plastic surgeon will opt for a face-lifting.

Most often, however, other «lightweight» methods are used before or as a complement to these techniques.

Abrasion and filling methods are used to act on other signs of skin aging, wrinkles and lines.

- The former are used to eliminate the superficial layer of the epidermis marked by lines. There is a dual effect of this peeling, dermoabrasion and other «laser resurfacing».
  In addition to the flattening of skin relief obtained, the new skin regenerated in response to the treatment is smoother and finer.

- The latter involve filling lines and wrinkles, even adding volume by implanting injectable materials, resorbable or not. In the first case, i.e. resorbable, the products injected under facial wrinkles and lines (collagen, hyaluronic acid, polylactic acid, elastin) are progressively eliminated by the organism and the resulting correction lasts only 6 to 9 months.
  In the case of injectable non-resorbable products, the results last for several years.

In both cases, the material injected is either synthetic, or the patient's own adipose tissue taken from another site in the body (generally from the hips, abdomen or thighs). In this case, we speak of the filling technique.

It can be done under local anesthesia and has the advantage of using a natural product with no side effects (allergic reaction for example). Even though all these techniques have many advantages, they nevertheless involve surgery, are obviously invasive and are very costly.

Become younger without surgery

BASF Beauty Care Solutions has found inspiration in plastic surgery methods and proposes a patented*, non-invasive «filling» technique. It uses materials obtained with biotechnology that are perfectly bio-compatible to fill deep wrinkles.

Hyaluronic Filling Spheres™ are composed of a mixture of high and low molecular weight hyaluronic acids, perfectly accepted by skin tissue. They are incorporated in the oily phase of cosmetic formulations in a dehydrated form to better penetrate the skin.
When they are applied, they easily lodge in skin tissue depressions (wrinkles) where they progressively but rapidly rehydrate by taking up water that evaporates from the skin via the well known process of TransEpidermal Water Loss (TEWL) *(figure 1).*

This physical action leads to a volume change of Hyaluronic Filling Spheres™ in the epidermis that almost immediately smoothes deep wrinkles such as expression wrinkles, at the same time increasing the long-term hydration of the upper layers of the epidermis.

The strategy adopted when Hyaluronic Filling Spheres™ were being developed is based on three major complementary orientations:

1) **The choice of the bio-compatible materials with a very high water binding capacity**,

2) demonstration *in vitro* and *in vivo* of swelling properties in aqueous medium and on the skin,

3) The demonstration *in vitro* that there is no swelling in a standard formulation.

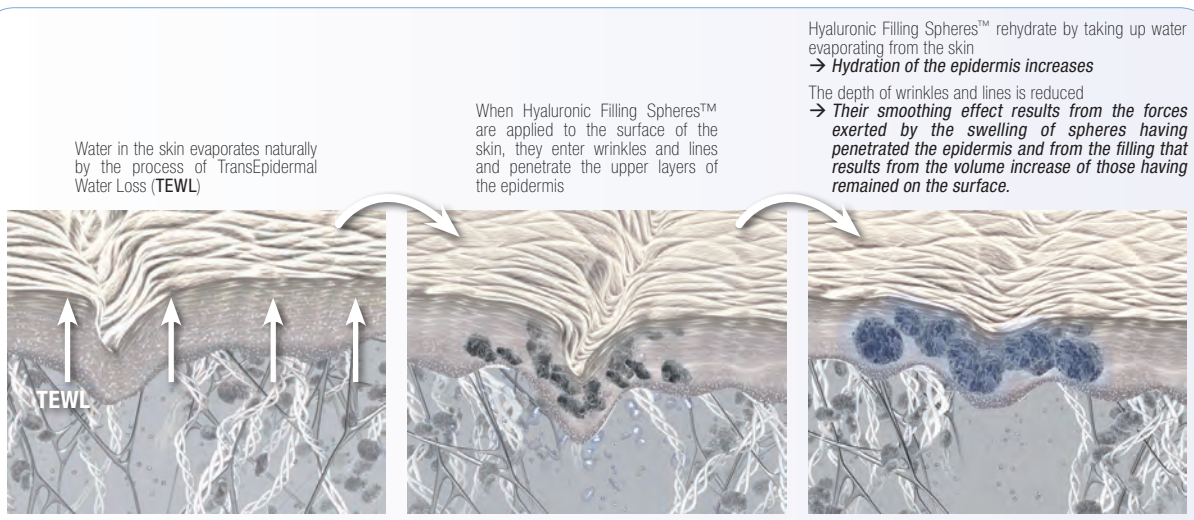


Figure 1: Mechanism of action of Hyaluronic Filling Spheres™.



Hyaluronic Filling Spheres™: a demonstrated efficacy

Choice of the biomaterial

For many years both dermatology and cosmetology have paid special attention to hyaluronic acid (HA) for its numerous qualities, in particular in the areas of healing, hydration and skin aging and hydration, notably as a wrinkle-filling product.

Discovery

In 1939 Meyer and Palmer identified a viscous component in the vitreous humor of bovine eyes.
After the chemical determination of its primary structure, it was placed in the large family of polysaccharides and was called hyaluronic acid, from the Greek «hyalos» (glass) because of its vitreous appearance.

Since the 1950s, a large number of studies have shown the efficacy of hyaluronic acid in arthritis. In the 1970s, the compound underwent overnight success in the field of ophthalmology, as a complement to cataract surgery, where it was used as a «visco-supplementation», factor[2].

The 2000s will see it sit in its rightful place in the area of plastic surgery as a wrinkle-filling material.

Characteristics and physiological properties

HA is an acid in name only. It is in fact a glycosaminoglycan (GAG), a high molecular weight polysaccharide (several hundred thousand to several million Daltons). It is a major component of extracellular matrices of which the skin is the principal reservoir, since it contains half of the total in the body, estimated at about 15 g. HA or hyaluronan is a linear polymer constructed by the repetition of 500 to 1000 disaccharide units having the structure [D-glucuronic acid (1-β-3) N-acetyl-D-glucosamine (1-β-4)]n *(figure 2)* and synthesized primarily by fibroblasts and keratinocytes[3].
In association with the other substances of the intercellular cement, it fills the spaces separating collagen and elastin fibers.
HA is known for its powerful hygroscopic properties: it is capable of binding a large quantity of water molecules – explaining its nickname of «molecular sponge» - forming extremely viscous solutions whose lubricating and spreading properties produce an excellent hydrating effect. In the skin it plays the physiological role of a reservoir by retaining water in the dermis and a mechanical role of damper that improves biomechanical properties of the skin such as suppleness and firmness:

- in the joints, its essential viscoelastic capacity acts in the synovial fluid as a shock absorber and lubricant[2],
- in tissues it plays a major role in homeostasis of the skin; it controls ionic fluxes and acts as a barrier against rapid changes in water contents[4],
- in the umbilical cord it protects arteries and increases elasticity,
- in the skin and cartilage, hyaluronic acid forms a matrix tightly bound to water molecules and collagen fibrils; this property is occasionally used to prepare biomaterials,
- in terms of healing wounds, the presence of an extracellular matrix enriched in hyaluronate is

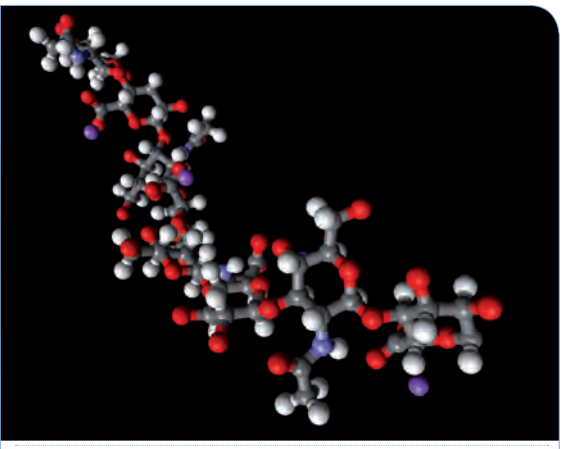


Figure 2: Molecular structure of hyaluronic acid

characteristic of the first stages of tissue reconstruction,

- it facilitates fibroblast migration in tissue cultures and the healing process.

Regulation of HA

Hyaluronan synthase is localized on the inner side of the fibroblast plasma membrane and is stimulated only by the presence of hyaluronidase: it is the degradation of HA that stimulates its biosynthesis.
Hyaluronic acid can undergo partial degradation on the surface of the skin under the effect of hyaluronidase and singlet oxygen. The oligosaccharides produced also have important properties such as the capacity to trap free radicals or trigger the synthesis of collagen (types I and VIII).

AH et vieillissement cutané

The quantity of HA decreases in the dermis with age[5] a deterioration of tissue quality and repair processes. As a result, all the above-mentioned properties have a direct application to skin aging.
For example, its viscoelastic properties enable it to play the role of a damper, thereby avoiding excessive ruptures of the skin, responsible for wrinkles.
Its support capacity via adhesion molecules limits the separation of cells from their extracellular matrix, even anchoring fibers from the epidermis to the dermis.
Its hydrating and hygroscopic properties enable water and electrolytes essential for the life of cells to be retained[6].

Description

Hyaluronic Filling Spheres™ are incorporated in a totally anhydrous oily support and appear as a white to off-white, translucent, viscous liquid.
In their dehydrated form, Hyaluronic Filling Spheres™ are a white powder *(photo 1)* composed of a mixture of high molecular weight HA.
Initial studies conducted when Hyaluronic Filling Spheres™ were being developed showed the extraordinary capacity of these structures to increase their volume in response to hydration *(photo 2).*

Thus, after mixing 0.4 g of dehydrated Hyaluronic Filling Spheres™ with 9.5 g of 1% citrate buffer, pH 7, with rapid mixing on a vortex mixer followed by centrifugation, the result was immediate. In contact with the buffer; the volume of the biomaterials used was considerably increased and the viscosity of the preparation became relatively high *(photo 3).*
Following this initial work, tests were conducted to determine the swelling of Hyaluronic Filling Spheres™ after hydration by light and scanning electron microscopy. Tests with light microscopy involved visualizing the spheres in aqueous medium and strippings prepared after applying Hyaluronic Filling Spheres™ to the skin.



Commercial sample
of Hyaluronic Filling Spheres™

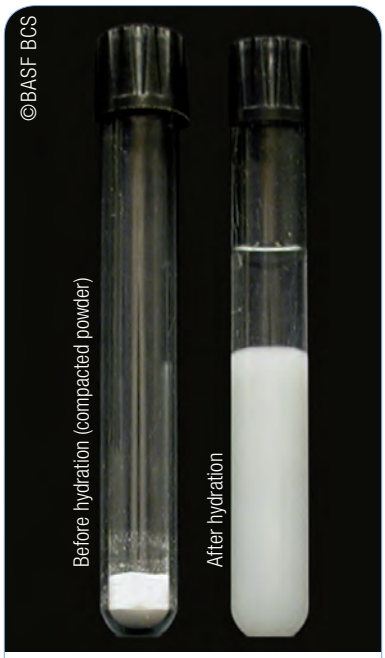


Photo 2: Visualization *in vitro* of the swelling of Hyaluronic Filling Spheres™ in aqueous medium.



Photo 1: Dehydrated Hyaluronic Filling Spheres™.



Photo 3: Increased volume of Hyaluronic Filling Spheres™ in contact with aqueous buffer

OBJECTIVE

The aim of this study is to use optical and confocal microscopy to visualize and measure the ability of dehydrated Hyaluronic Filling Spheres™ to soak up water and swell.

RESULTS

Optical microscopy

When Hyaluronic Filling Spheres™ were observed as a freeze-dried powder, they appeared small and dense under the optical microscope.

After re-hydration, they had soaked up large quantities of water and appeared larger. However, it was impossible to estimate their thickness and volume using optical microscopy. Further investigations required confocal microscopy techniques *(figure 4)*.

Confocal microscopy

Using confocal microscopy image analysis, the volume of 11 dehydrated Hyaluronic Filling Spheres™ was calculated and compared to that of 11 re-hydrated Hyaluronic Filling Spheres™. The same calculation formula was used for both dehydrated and rehydrated forms *(figure 5)*. These results clearly show that after water immersion, the Hyaluronic Filling Spheres™ enlarged in all three dimensions as demonstrated by the x, y and z values.

The overall swelling factor calculated as the volume rate between re-hydrated and dehydrated Hyaluronic Filling Spheres™ increased by x **30.44.** In other words, after only 10 minutes of re-hydration, Hyaluronic Filling Spheres™ had absorbed water as much as 30 times their initial volume.

CONCLUSION

As observed by optical microscopy and calculated thanks to confocal microscopic technique, Hyaluronic Filling Spheres™ could rapidly swell by soaking up a large quantity of water.

Dimensions	x (µm)	y (µm)	z (µm)	xyz (µm3)	Volume±SD (µm³)
Dehydrated Hyaluronic Filling Spheres™	10.34	10.39	8.06	902	3777±1532
Rehydrated Hyaluronic Filling Spheres™	35.81	37.11	19.91	27464	114984±50423***
Swelling factor (Re-hydrated spheres' volume/ Dehydrated spheres' volume)					30.44

Table 1: Measurement of the swelling factor of Hyaluronic filling Spheres™.

****: Results are expressed as means and means±SD (***: p<0.001, statistically significant vs. dehydrated spheres)*

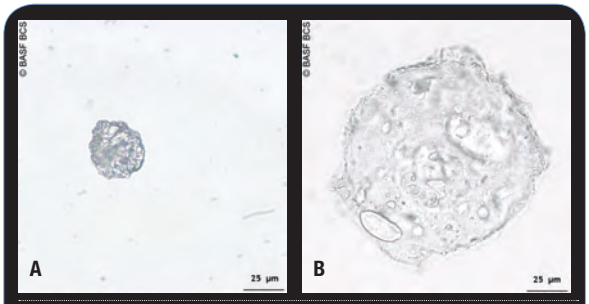


Figure 4: Optical microscopic observation of dehydrated (**A**) and rehydrated (**B**) Hyaluronic Filling Spheres™ (magnification, x40).

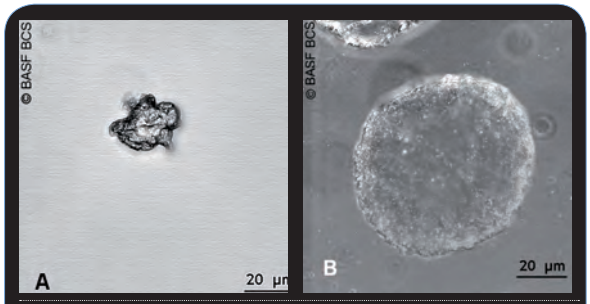


Figure 5: Confocal microscopic observation of dehydrated (**A**) and rehydrated (**B**) Hyaluronic Filling Spheres™ (magnification, x63).

MATERIALS&METHODS

Tested products

Hyaluronic Filling Spheres™ were observed as dehydrated (freeze-dried) and re-hydrated spheres (10 minutes in deionized water).

Optical microscopy

Observations were performed using an optical microscope Axioskop 2+ *(Zeiss, Germany)* with a Plan-Neofluar objective (magnification, x40, numerical aperture 0.75) and brightfield illumination. Pictures were taken with a digital camera Dxm1200f *(Nikon, France)* and analyzed using Lucia software *(Nikon, France)*.

Confocal microscopy

Confocal observations were carried-out using a Zeiss LSM 510 microscope with a Plan-Neofluar objective (magnification x 63, numerical aperture 1.4 in oil) and Ar and He/Ne laser excitation (λ=488 and 543 nm respectively).

Images acquisitions were performed using Zeiss LSM Meta 3.2 software associated with a Z-stack protocol to determine the thickness, width and length of each observed microsphere.

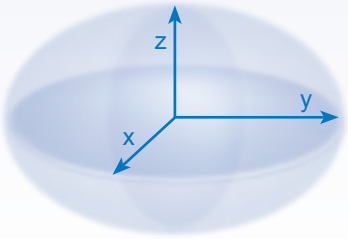
Single projection was achieved with a 3D/ projection module.

Measurement principle

Microspheres can be modelled as ovoid objects whose volume can be evaluated using the following formula:

V (μm^3) = $4/3*\pi*$ (**x*****y*****z**) with x, y and z (µm) corresponding to ½ microsphere width, length and thickness respectively.

The statistical significance between dehydrated and re-hydrated spheres was evaluated using Mann-Whitney U test *(***: p<0.001)*.



OBJECTIVE

The aim of this study is to evaluate the enzymatic hydrolysis of Hyaluronic Filling Spheres™ using an *in vitro* test involving two proteases similar to those currently found in the epidermis. The hydrolysis of the spheres is then observed using optical microscopy.

RESULTS

Immediately after incubation, intact Hyaluronic Filling Spheres™ are visible *(figure 6)*.

After 24 hours many Hyaluronic Filling Spheres™ incubated with proteases are hydrolyzed and small membrane fragments are observed. Conversely, Hyaluronic Filling Spheres™ incubated without proteases are intact after 24 hours *(figure 7)*.

CONCLUSION

The results of this in vitro study show that Hyaluronic Filling Spheres™ are hydrolyzed by skin proteases demonstrating the biodegradable nature of Hyaluronic Filling Spheres™ technology.

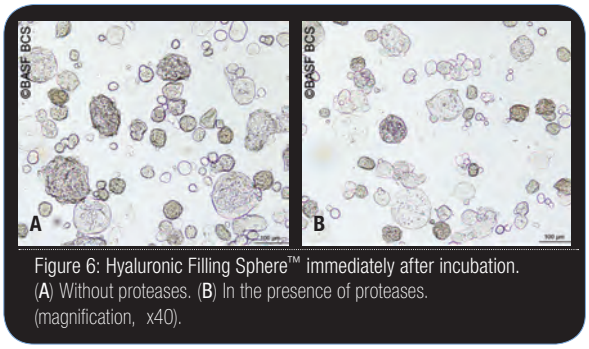


Figure 6: Hyaluronic Filling Sphere™ immediately after incubation. (**A**) Without proteases. (**B**) In the presence of proteases. (magnification, x40).

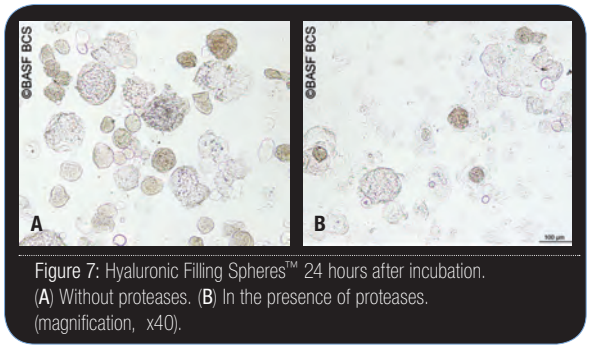


Figure 7: Hyaluronic Filling Spheres™ 24 hours after incubation. (**A**) Without proteases. (**B**) In the presence of proteases. (magnification, x40).

MATERIALS&METHODS

Proteases selection

A bibliographic research enabled us to identify two major proteases usually found in the epidermis: the *Stratum corneum* tryptic enzyme (SCTE) and the *Stratum corneum* chymotryptic enzyme (SCCE).

SCTE and SCCE are two serine proteases involved in skin normal desquamation process (Caubet C. and coll., *J. Invest. Dermatol.*, 2004, 122(5):1235-44).

Neither of them being commercially available, we selected bovine pancreatic trypsin and α-chymotrypsin for this experiment. Indeed, according to the protein sequence database ExPASY, bovine pancreatic trypsin and α-chymotrypsin active sites are highly homologous to those of human SCTE and SCCE (100% and 67% for SCTE and trypsin; 83% and 75% for SCCE and α-chymotrypsin).

Experimental procedure

0.3g of lyophilized Hyaluronic Filling Spheres™ were diluted in 100g of deionized water and incubated for 10 minutes to obtain maximum swollen spheres.

The solution was then incubated at pH 6 with high amounts of enzymes: 3.342.740 enzymatic units of trypsin and 397.783 enzymatic units of α-chymotrypsin.

Observations were performed immediately after incubation, then after 24 hours using an Axioskop2+ optical microscope *(Zeiss, Germany)* with an A-Plan objective (magnification x10, numerical aperture 0.25). Pictures were taken using a Dxm1200f camera and analyzed using Lucia software *(Nikon, France)*.

OBJECTIVE

The aim of this *in vivo* study is to follow the penetration of Hyaluronic Filling Spheres™ after one single application on the skin. Successive tape strippings are performed on epidermal superficial layers treated with Hyaluronic Filling Spheres™ and visualized using optical microscopy.

RESULTS

Immediate tape strippings

Pictures from the different strippings sampled 10 minutes after application of the control product show visible corneocytes. Some skin furrows between corneocytes can also be distinguished *(figure 8)*.

Pictures of successive strippings from area 2 (Hyaluronic Filling Spheres™ treated area) show individualized blue spheres (= rehydrated spheres) between corneocytes.

In these conditions, i.e. immediately after a single application, some rehydrated Hyaluronic Filling Spheres™ can be distinctly observed up to the second stripping *(figure 9)*.

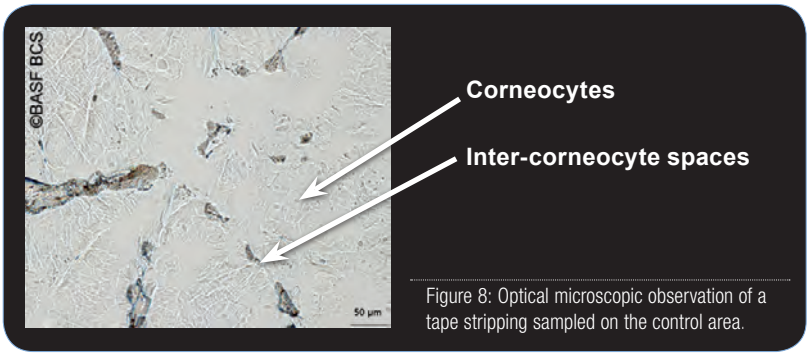


Figure 8: Optical microscopic observation of a tape stripping sampled on the control area.

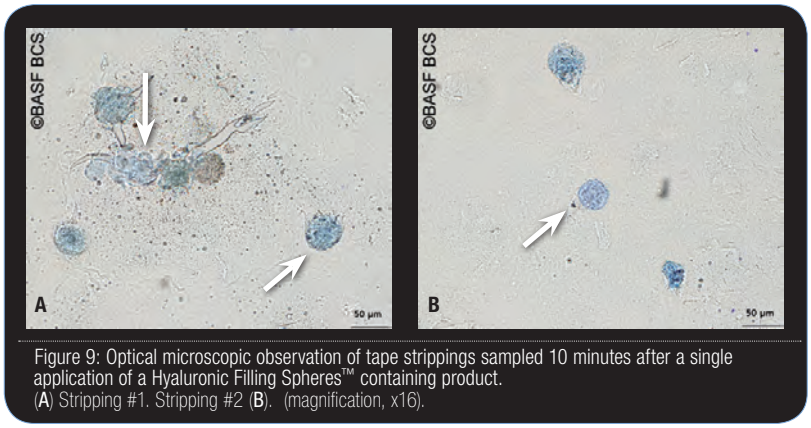


Figure 9: Optical microscopic observation of tape strippings sampled 10 minutes after a single application of a Hyaluronic Filling Spheres™ containing product.
(**A**) Stripping #1. Stripping #2 (**B**). (magnification, x16).

MATERIALS&METHODS

Tested products

Oily medium whether containing Hyaluronic Filling Spheres™ (trial) or not (control).

Applications

10 µl of each sample were applied once on a human volunteer's forearm skin and massaged for 30 seconds.

Four different areas of 4 cm² each were defined on the forearm as follows:

Area 1: Control medium treated-area sampled 10 minutes after application

Area 2: Area treated with Hyaluronic Filling Spheres™ and sampled 10 minutes after application

Area 3: Area treated with control medium and sampled 5 hours after application

Area 4: Area treated with Hyaluronic Filling Spheres™ and sampled 5 hours after application.

Tape strippings

10 minutes after the massage, D-Squam tapes (Monaderm, Monaco) were applied on areas 1 and 2 using a disk applicator *(Monaderm, Monaco)* with a standardized pressure exerted for 20 seconds. The tapes were then removed and deposited on their non-adhesive side on a glass slide before microscopic observation.

In order to follow spheres' presence throughout the epidermal outermost layers, six strippings were successively performed on the treated areas and observed.

After 5 hours, the same procedure was performed on areas 3 and 4.

Visualization

Observations were performed using an optical microscope Axioskop 2+ *(Zeiss, Germany)* with Plan-Neofluar objective (magnification x16, numerical aperture 0.5 in water) with brightfield illumination in 0.05% methylene blue immersion.

Pictures were taken using a digital camera Dxm1200f *(Nikon, France)* and analyzed using Lucia software *(Nikon, France)*.

Tape strippings sampled 5 hours after application

Tape strippings performed on areas 3 and 4 (untreated and treated areas respectively, both 5 hours after application) clearly show re-hydrated Hyaluronic Filling Spheres™ up to the sixth stripping. Indeed, spheres density gradually decreases with the increasing number of strippings *(figure 10)*.

CONCLUSION

This study has shown Hyaluronic Filling Spheres™ ability to penetrate the different epidermal layers.

First, when dehydrated, their small size allows them to easily penetrate the epidermal upper layers after application on skin surface.

Then, 10 minutes after application, Hyaluronic Filling Spheres™ start to re-hydrate by taking up water evaporating from skin depth and are distinctly visible under optical microscope up to the second stripping.

Five hours after the single application, re-hydrated Hyaluronic Filling Spheres™ can easily be seen at least up to the sixth stripping from the same skin area.

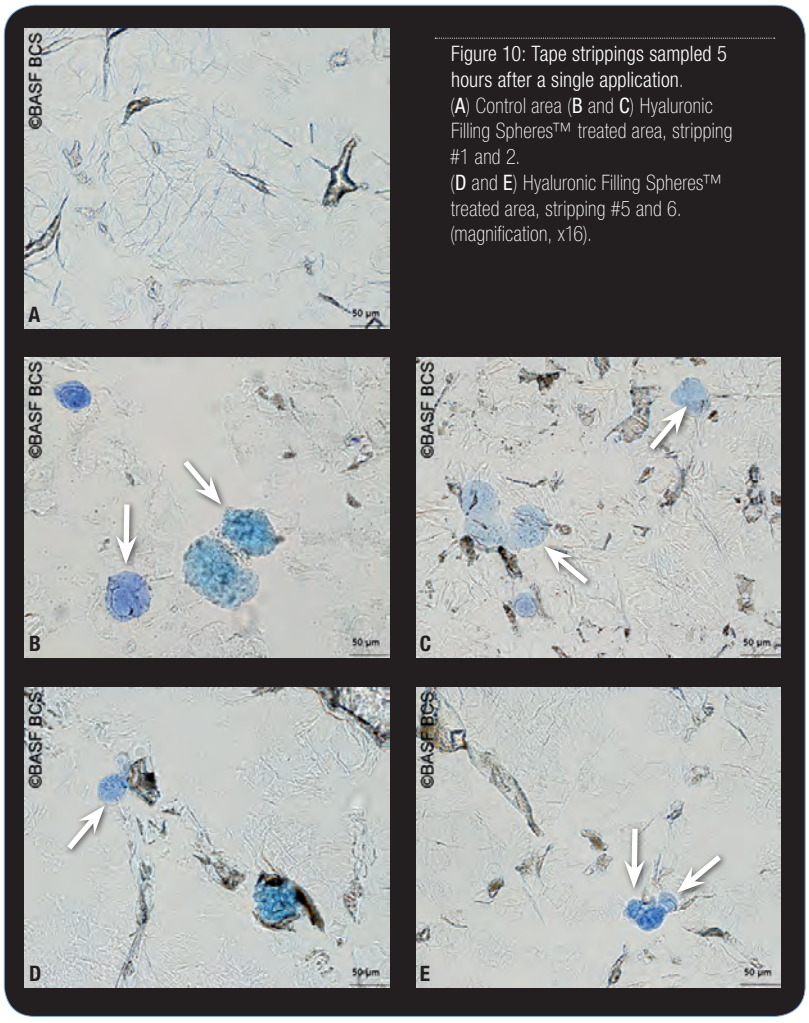


Figure 10: Tape strippings sampled 5 hours after a single application.
(**A**) Control area (**B** and **C**) Hyaluronic Filling Spheres™ treated area, stripping #1 and 2.
(**D** and **E**) Hyaluronic Filling Spheres™ treated area, stripping #5 and 6.
(magnification, x16).

OBJECTIVE

The aim of this in vivo study was to use light microscopy photos to show the swelling power of Hyaluronic Filling Spheres™ on the skin.

RESULTS

Compared to the photo of the control (*figure 11**A***), the photos in *11**B**, 5**C*** and *5**D*** illustrate the capacity of Hyaluronic Filling Spheres™ to swell by taking up and retaining water that evaporates from the skin via the phenomenon of TEWL.

This *in vivo* study on the surface of forearm skin showed that the spheres, initially dehydrated, «swelled» in only 1 hour and that this volume increase persisted for at least 5 hours.

At 5 hours, it is noteworthy that most of the swollen spheres were localized in skin folds which thus appeared darker.

CONCLUSION

This study of the surface of the skin showed that Hyaluronic Filling Spheres™ could also «swell» *in vivo,* by taking up water that evaporated from the superficial layers of the skin via the phenomenon of TEWL.
These «mini-traps» can penetrate the upper layers of the epidermis to become rapidly transformed into «mini-reservoirs of water».

After having shown that Hyaluronic Filling Spheres™ swell both *in vitro* and *in vivo* in contact with water, it was just as important to demonstrate that the spheres did not swell in cosmetic formulations, i.e. before their application on the skin.

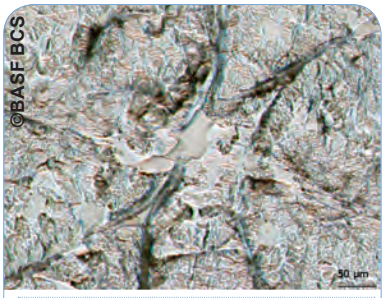


Figure 11A: Stripping before application (control).

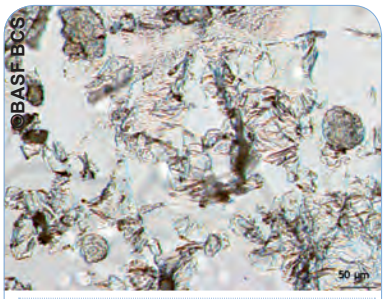


Figure 11B: Hyaluronic Filling Spheres™ 1h after application

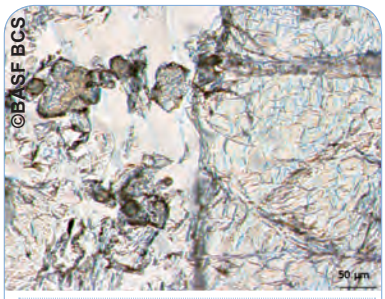


Figure 11C: Hyaluronic Filling Spheres™ 3h after application

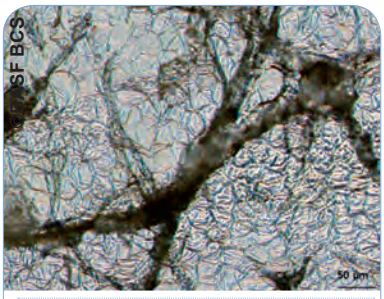


Figure 11D: Hyaluronic Filling Spheres™ 5h after application

MATERIALS&METHODS

Dehydrated Hyaluronic Filling Spheres™ were added to a support medium (oily mixture of octyl palmitate and dimethylsilylate silica with preservatives) at the concentration of 5% and 10 µl of the mixture were applied to four skin zones on the inside of the forearms of a volunteer.

- Zone 1: before application (control)
- Zone 2: T 1h after application
- Zone 3: T 3h after application
- Zone 4: T 5h after application

After gently massaging to disperse the spheres on the surface of the skin and after a delay of 5 minutes, D-SQUAME® *(Monaderm, Monaco)* adhesive was applied to each zone with an automatic applicator that ensured an identical pressure on the four sampling zones *(Monaderm piston, Monaco).*
After t=1h, t=3h and t=5h the adhesives were recovered and visualized under light microscopy using a direct illumination microscope over a light background *(Axioskop2 Plus, Zeiss, Germany)* with a x16 objective coupled with an acquisition system composed of a Dxm1200f camera and a Lucia image analyzer *(Nikon, France).*

Light microscopy visualization of the lack of swelling of Hyaluronic Filling Spheres™ when in contact with different oily supports

STUDY IN A FORMULATION

OBJECTIVE

The aim of this *in vitro* study was to use light microscopy photos to show that Hyaluronic Filling Spheres™ incorporated in a standard cosmetic formulation do not swell.

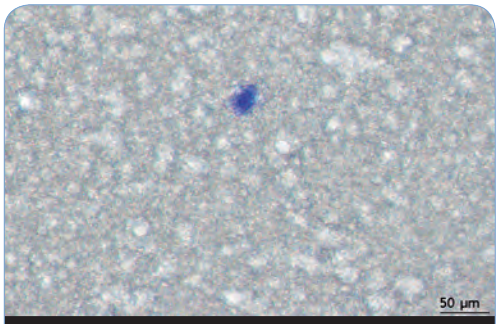


Figure 12A: Hyaluronic Filling Spheres™ incorporated in oily phase of an O/W emulsion = no swelling (x16).

RESULTS

Visualization of the test of non-swelling of Hyaluronic Filling Spheres™ in a standard cosmetic formulation

As shown in the photos obtained in this study, when Hyaluronic Filling Spheres™ are added to the oily phase of an O/W emulsion, no swelling was observed *(figure 12**A**)*.

In an oily medium with the absence of water, the majority of the spheres remain dehydrated.

Conversely, as shown by the photos of the Oil/(water + spheres) emulsion used as the positive control, when the spheres were added to the aqueous phase, in direct contact with water, they swelled immediately *(figures 6**B** et 6**C**)*.

Au delà de leur effet sur l'hydratation des couches supérieures de l'épiderme, les Hyaluronic Filling Spheres™ devaient aussi être capables en «gonflant», de «lisser» la surface du tissu cutané.

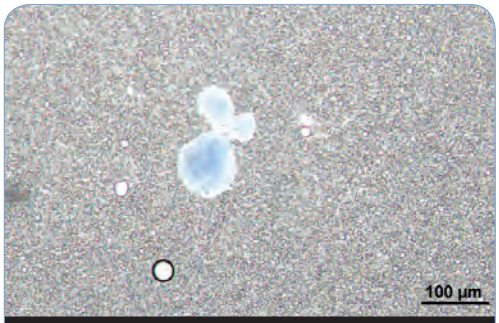


Figure 12B: Hyaluronic Filling Spheres™ incorporated in aqueous phase of an O/W emulsion = swelling (x10).

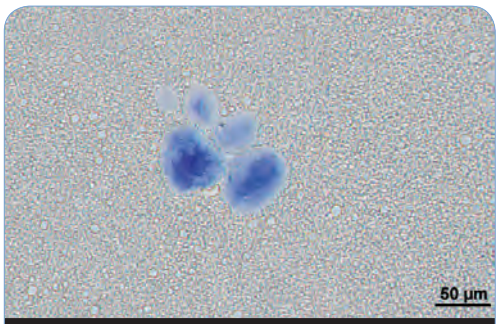


Figure 12C: Hyaluronic Filling Spheres™ incorporated in aqueous phase of an O/W emulsion = swelling (x16).

CONCLUSION

These two studies demonstrate that in the absence of water, Hyaluronic Filling Spheres™ cannot swell.

The spheres thus remain in their dehydrated form in both the 100% oily support medium and the standard O/W cosmetic formulation until they are applied to the skin, provided they are added to the oily phase of the emulsion.

MATERIALS&METHODS

An O/W formulation (26% oily phase/74% aqueous) was prepared in which 0.05% (w/w) methylene blue was dissolved in the aqueous phase and 2% hyaluronic was added to the oily phase with mixing.

The oily phase was emulsified with the aqueous phase with energetic mixing and heating and the emulsion (oil + spheres/water) was slowly cooled with continued mixing.

In parallel, Hyaluronic Filling Spheres™ were also added with methylene blue to the aqueous phase of an emulsion used as a positive control [oil/(water + spheres)]. Light microscopy observation was conducted by preparing slides with the same thickness of cream obtained, using a piston *(Monaderm, Monaco)* by depositing 50 µg of the preparation.

Observation was with an up-right light microscope against a light background *(Axioskop2 Plus Zeiss, Germany)* using x16 and x40 water immersion objectives, coupled with an acquisition system composed of a Dxm1200f camera and a Lucia image analyzer *(Nikon, France)*.

OBJECTIVE

Two different clinical studies were performed to measure the ability of Hyaluronic Filling Spheres™ to quickly reduce the appearance of wrinkles after a single application. First, the efficacy of Hyaluronic Filling Spheres™ *in vivo* was assessed by clinical evaluation. Second, crow's feet wrinkles were analyzed using interference microscopy.

RESULTS

Clinical evaluation

Compared to T0, a significant decrease in the appearance of the forehead expression wrinkles was noted 1 hour and 6 hours after the product application, for both tested products *(figure 13)*.

However, this decrease is bigger for the cream containing Hyaluronic Filling Spheres™ at 2% than the placebo.

	After 1 hour	After 6 hours
Hyaluronic Filling Spheres™ 2%	- 7.5%, p <0.001	- 12%, p <0.001
Placebo	- 4.5%, p <0.05	- 6%, p <0.05

Compared to the non treated wrinkles control and to the different times of the study, the treated wrinkles with the cream containing Hyaluronic Filling Spheres™ at 2% were appeared less depth *(figure 14)*:

A T1 hour: Δ = 7.5 %, p <0.001
A T6 hours: Δ = 12 %, p <0.001

However, no significant difference was noted between placebo and control.

Six hours after single application, the cream containing Hyaluronic Filling Spheres™ at 2% is statistically more efficient to decrease the forehead wrinkles appearance than the placebo (Δ = 6 %, p <0.05; analysis on the differences ΔTx – T0) *(figure 15)*.

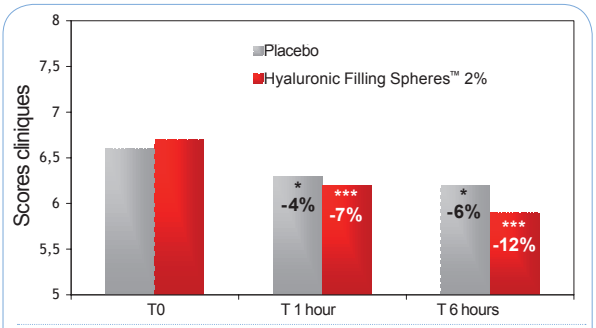


Figure 13: Clinical evaluation of the appearance of expression wrinkles. Time effect (n=26).

**: statistically significant compared to the T0 (p<0.05)*
****: statistically significant compared to the T0 (p<0.001)*

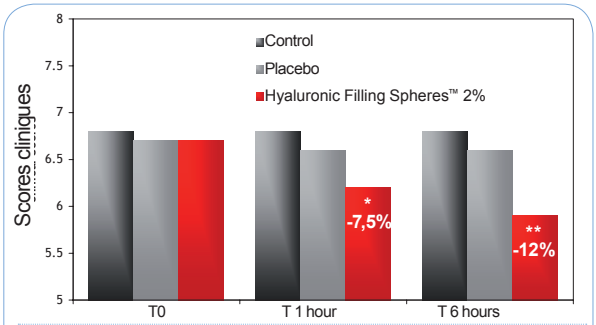


Figure 14: Clinical evaluation of the appearance of expression wrinkles. Product effect versus control (n=13).

**: statistically significant compared to the control (p<0.05)*
***: statistically significant compared to the control (p<0.01)*

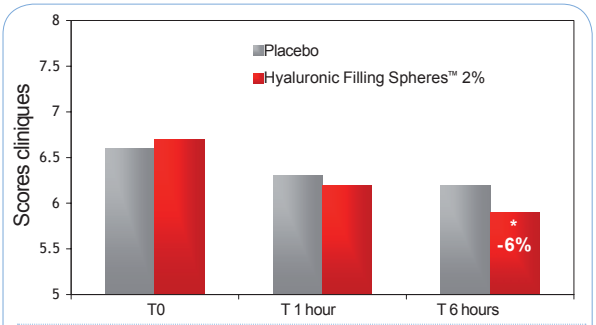


Figure 15: Clinical evaluation of the appearance of expression wrinkles. Product effect versus placebo (n=26).

**: statistically significant compared to the placebo (p<0.05)*

Appraisal of the efficacy of the test products by the volunteers (n = 26)

From the volunteer's answers obtained, the following positive responses were revealed:

	Placebo	Hyaluronic Filling Spheres™
The skin is smoothed	62%	77%
The forehead expression wrinkles are softened	31%	54%



MATERIALS&METHODS

Clinical evaluation

Volunteers

Twenty six female adult volunteers (41 to 55 years old), presenting expression wrinkles on the forehead were recruited to participate in this clinical study.

Tested products

- An O/W formula containing Hyaluronic Filling Spheres™ at 2%
- A Placebo = the same O/W formula without Hyaluronic Filling Spheres™

The distribution of each test article was randomized and in cross over (each volunteer would have tested both products with an 8 days wash out period between). Tested products were applied on the whole face skin, by the volunteer herself, under the normal conditions of use instead of what she normally uses.

Methodology

For each treatment and for each volunteer (n=26), clinical scores are noted on the forehead expression wrinkles:

- at T0 before application,
- at T1 and 6 hours after one single application.
  For thirteen subjects among the panel, measures were done without any treatment = control.
- evaluation by a trained expert (clinical scoring in nine points (1 to 9)) of the "appearance of the wrinkles".
- lAssessment of the efficacy and cosmetic qualities of the test article by the volunteers, on the basis of a succinct questionnaire.
- For five subjects, macrophotographs of the wrinkles'forehead were taken with a digital camera NIKON (D100) before and after application of Hyaluronic Filling Spheres™ at 2%.

Instrumental study

In this second study, the anti-wrinkle efficacy of a cosmetic formula containing 2% of Hyaluronic Filling Spheres™ was compared to that of a placebo 6 hours only after a single application.

Silflo™ silicone replicas were taken from the volunteers' crow's feet before and 6 hours after application of the products. Then, all the replicas are analyzed using a white light interference microscope. The efficacy of the products is evaluated over time and a comparison between products is also performed.

The results presented in *Fig. 16* show that Hyaluronic Filling Spheres™ allowed for a significant **7.2%** reduction in wrinkle angle ($p<0.05$). This result was better than that obtained with the placebo for which a -6.7% reduction was observed ($p<0.05$).

Regarding wrinkle depth, the treatment with Hyaluronic Filling Spheres™ led to a **-9.4%** reduction *(Fig. 17)*. This result was statistically significant compared to the placebo ($p<0.001$) which on the contrary significantly increased wrinkle depth by +30% ($p<0.01$).

Likewise, Hyaluronic Filling Spheres™ allowed reducing wrinkle width by **-5.1%** *(Fig.18)*. Same as previously, this result was statistically significant compared to the placebo ($p<0.001$) which significantly increased wrinkle width by +26% ($p<0.001$).

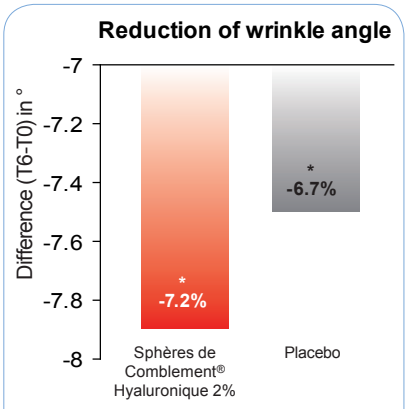


Figure 16: Variation in wrinkle angle between T0 and T6 hours.

**: statistically significant vs. T0, p<0.05*

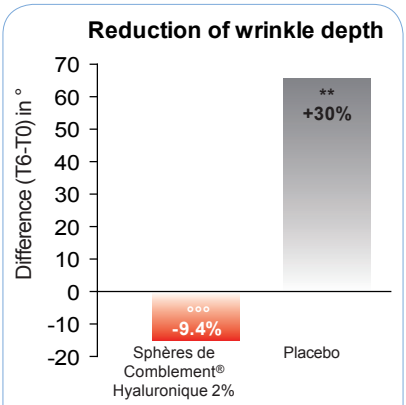


Figure 17: Variation in wrinkle depth between T0 and T6 hours.

***: statistically significant vs. T0, p<0.01*
°°°: statistically significant vs. placebo, p<0.001

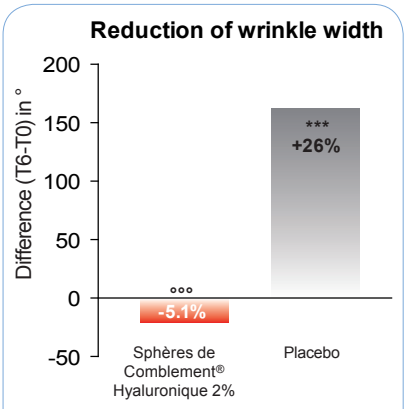


Figure 18: Variation in wrinkle width between T0 and T6 hours.

****: statistically significant vs. T0, p<0.001*
°°°: statistically significant vs. placebo, p<0.001

Figure 19 shows an example of a crow's foot before and after treatment with Hyaluronic Filling Spheres™. A visible reduction of wrinkle width can be observed.

Moreover, it can be clearly seen that the overall wrinkle relief is smoothed as shown by the disappearance of the red areas representing the maximum wrinkle height on the top right corner.

The surface between the two wrinkles also appears to have been plumped as shown by the disappearance of the blue color representing the deep areas in the middle.

Actually, the whole relief seems to have been lifted up and smoothed.

As far as the placebo is concerned, no improvement can be seen *(Fig. 20)*.

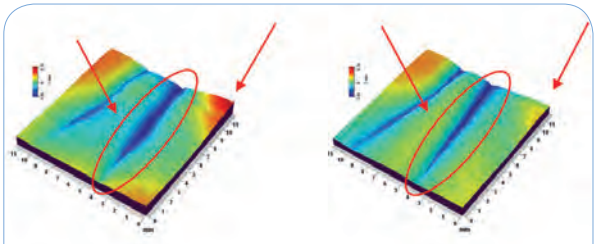


Figure 19: Example of reconstructed 3D image obtained from a crow's foot treated with Hyaluronic Filling Spheres™.

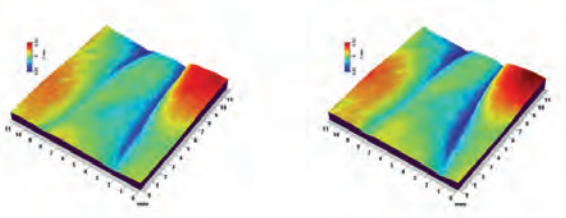


Figure 20: Example of reconstructed 3D image obtained from a crow's foot treated with placebo.

Fig. 21 shows illustrative pictures of a volunteer's crow's foot for which the reductions measured in wrinkle angle, depth and width were of **-10%**, **-3%** and **-11%** respectively.



Figure 21: Illustrative picture of a volunteer's crow before and 6 hours after treatment with Hyaluronic Filling Spheres™.

CONCLUSION

This double *in vivo* study allowed us to demonstrate the efficacy of Hyaluronic Filling Spheres™ at quickly smoothing wrinkles. First, the clinical evaluation highlighted that 6 hours only after a single application, the expression wrinkles of the volunteers treated with Hyaluronic Filling Spheres™ were significantly reduced compared to both baseline and placebo.

Interestingly, the instrumental study showed comparable results. Within 6 hours, the angle, width and depth of crow's feet wrinkles were significantly improved compared to placebo.

Thus, the *in vitro* results were confirmed *in vivo*: after having penetrated the upper layers of the epidermis, Hyaluronic Filling Spheres™ take up the evaporating water, swell, and provide quick visible wrinkle reduction.

MATERIALS&METHODS

Model

20 female volunteers aged 57 to 66 were recruited based on specific criteria such as moderate to severe wrinkles on the crow's feet.

Products tested

The same water in oil (W/O) cosmetic formula was tested containing either:

- 2% of Hyaluronic Filling Spheres™ or
- No spheres (placebo).

Protocol

All measurements were performed in the lab after a 15-minute rest period in a room with controlled temperature and humidity (21±1°C, HR 40±20%). The volunteers came to the lab without any cosmetic product on the face.

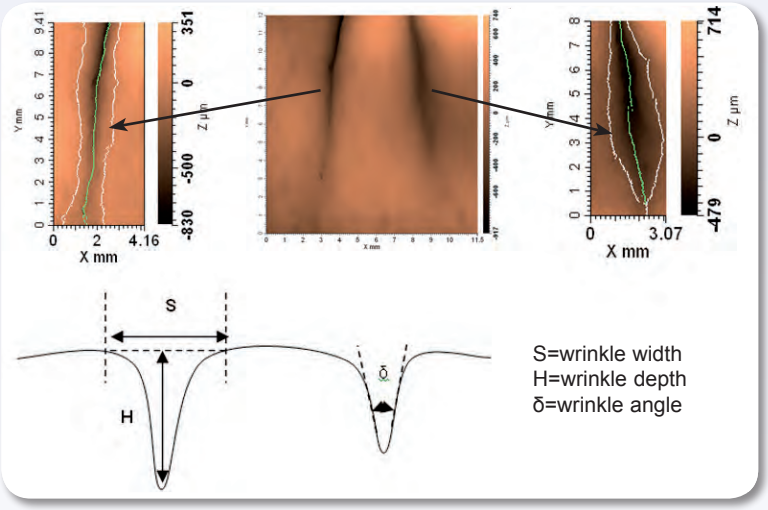
Silicone prints

A reference replica was first taken from both volunteers' crow's feet before application of the products.

The products to be tested were then applied by the technician. The formula containing Hyaluronic Filling Spheres™ was applied to one side of the face and the placebo was applied to the other side of the face after randomization. The volunteers then waited in a room, at the lab, for 6 hours before the second replica was taken following the same procedure as described above.

Illustrative pictures

Illustrative pictures of each volunteer's crow's foot were taken under cross-polarized light using a digital camera with standardized lighting before and after application of the products.



Replica analysis

All the replicas were analyzed using a white light interference microscope (WYKO Interferometer, AZ, USA) in vertical scanning mode (VSI). Data were recorded and analyzed using dedicated software (WYKO Vison™).

Principle of the measurement

The morphology of the wrinkle is investigated by quantifying the local geometrical parameters. The objective of the methodology is to find a way of locating the edge of the wrinkle automatically by assessing the brutal variation of the relief. For this, Each 512x512-pixel image is first scanned line by line and column by column in order to identify all the points that form the edge of the wrinkle. At the end of this screening, a threshold is fixed that will be the same for all the images to be analyzed and will serve as reference for the calculation of the wrinkle width (S), depth (H) and angle (γ). The figures above show an example of wrinkle detection and a schematic representation of the parameters studied. The white lines represent the limit of the wrinkle as defined by the threshold of relief variation. The green lines represent the maximum depth of the wrinkle.

Statistics

The statistical comparison over time (T6h vs. T0) for both Hyaluronic Filling Spheres™ and the placebo was assessed running Student's *t*-test for paired samples. The statistical comparison between products was assessed running Student's t-test for unpaired samples on the difference (T6h –T0). The significance threshold was set to 5% *($p<0.05$)* for all comparisons.



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Conclusion

Hyaluronic Filling Spheres™ bind large quantities of water in the superficial layers of the epidermis as a result of the presence of hyaluronic acid and its considerable hygroscopic properties.

By swelling almost immediately in contact with water, they increase the level of hydration of skin tissue while exerting the rapid physical action of filling deep wrinkles and expression wrinkles without the need for plastic surgery.

Specially developed by BASF Beauty Care Solutions to respond to the increasing demands for immediate cosmetic actions, Hyaluronic Filling Spheres™ today are an effective and highly interesting contribution to the long-term «anti-wrinkles» arsenal of the cosmetics industry.

Hyaluronic Filling Spheres™: an effective and alternative cosmetic solution to plastic surgery!

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Patent

Perrier E, Huc A, Buffevant C.
Polysaccharide wall microcapsules containing primary alcohol functions and compositions containing same.
(France: FR 2 688 422), (Europe: EP 0 630 287), (USA: US 5 562 924).

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